enable exchange of product information between PLM applications (for example, between a CAD application and a CAE application). They also enable exchange of product information between PLM applications and other enterprise applications such as ERP and CRM.

In a very relevant fashion, this middleware line of thinking is expanded upon by (Ben Khedher et al., 2011). In their work regarding different systems architectures for the implementation of an integrated MES+PLM they describe the use of a mediation system in web service architecture. As depicted in Figure 8, the proposed architecture uses data exchange based on internet technologies to help companies, especially expanded companies, to take advantage of opportunities generated by the Web Services. The concept of "web service" means an application (program or software system) which is designed to support interoperable machine-to-machine interactions over a network, according to the definition of W3C (Ben Khedher et al., 2011).

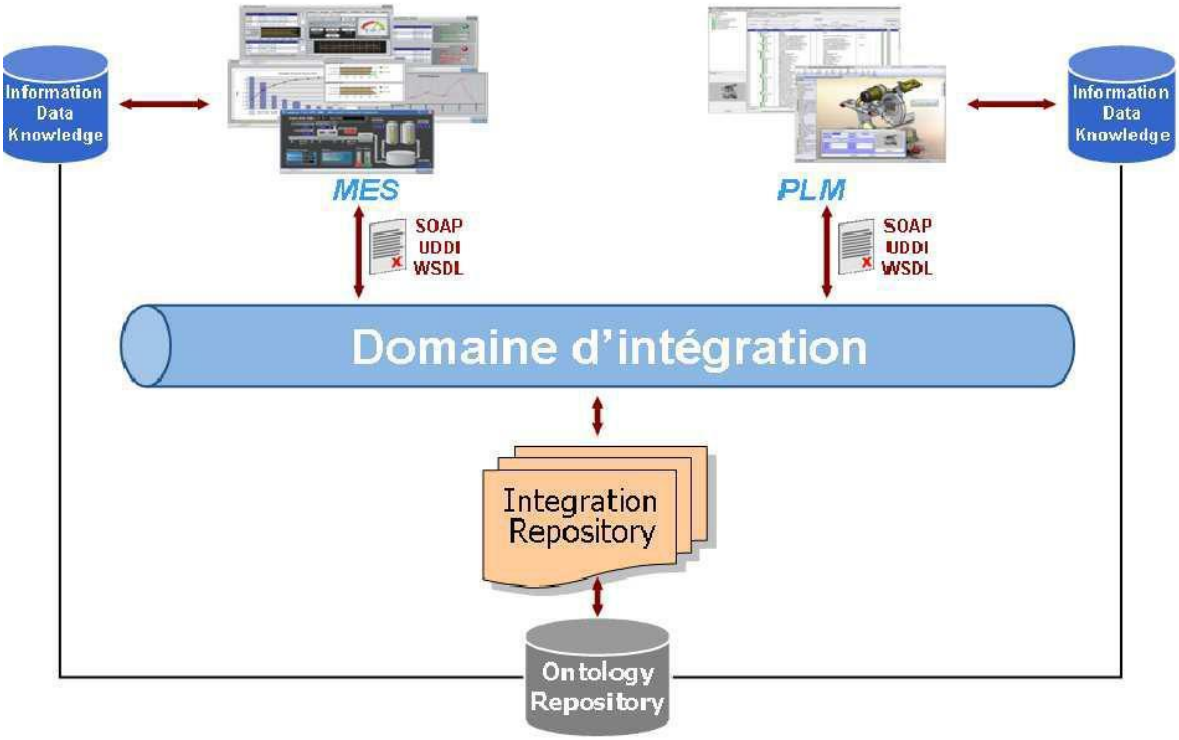


**Figure 8 Diagram of Web service architecture**
**(Adapted from Ben Khedher et al., 2011)**

The reason this expansion is so relevant from the perspective of this work is that the Odoo software works in a similar fashion through a similar web service architecture. In theory the Odoo software could act as the middleware working through the local network or hosted in the cloud and enacting the layer of integration that was previously mentioned.

實作 **PLM** 應 程式之間（例如 **CAD** 應 程式和 **CAE** 應 程式之間）的產品資訊交換。它們還 援 **PLM** 應 程式與其他企業應 程式（例如 **ERP** 和 **CRM**）之間的產品資訊交換。

**(Ben Khedher et al., 2011)** 以 常相關的 式擴展了這種中間件思路。在他們關於實施整合 **MES+PLM** 的不同系統架構的 作中，他們描述了中介系統在 **Web** 服務架構中的使 。如圖 **8** 所 ，所提出的架構使 基於互聯網技術的資料交換來幫助公司，特別是擴張型公司，利 **Web** 服務產 的機會。根據 **W3C** 的定義，「**Web** 服務」的概念是指旨在 援網路上可互通的機器對機器互動的應 程式（程式或軟體系統）（**Ben Khedher** 等，**2011**）。

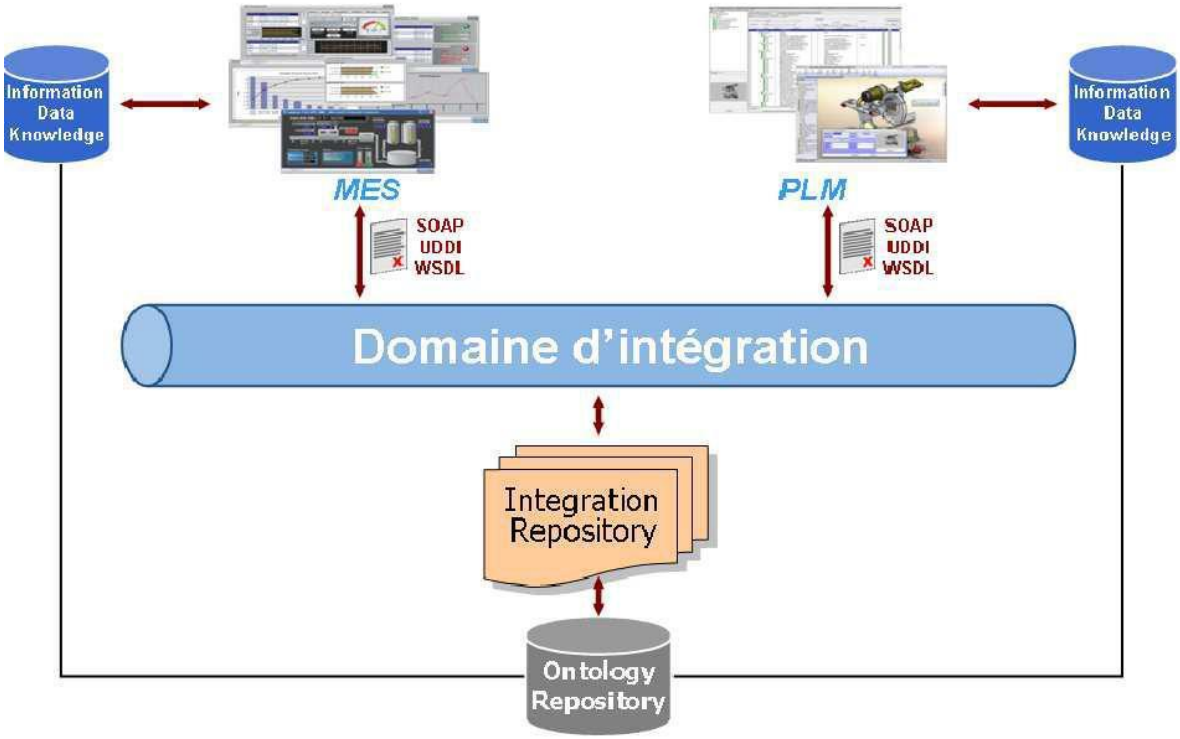


圖 **8 Web** 服務架構圖

（改編 **Ben Khedher** 等 ，**2011** 年）

從這項 作的 度來看，這種擴展如此相關的原因是 **Odoo** 軟體透過類似的 **Web** 服務架構以類似的 式運作。理論上，**Odoo** 軟體可以充當透過本地網路 作或託管在雲端中的中間件，並執 前 提到的整合層。

## 3.1. How would this integration look like in practical terms

As mentioned in CHAPTER 2 the main idea of PLM is to manage change in all processes related to the product, and it does so mainly through the use of virtualization. The word virtualization here denotes representation of item of the real world to the digital space and, as one can imagine, there are several levels of abstraction through which a real object or process can be represented. As consequence there is no exact consensus regarding PLM of how deep and/or detailed the virtual representation must be to serve its purpose.

In an ideal world that would be the lowest form of abstraction which, essentially, would come down to a digital twin as explained in the CHAPTER 2. This is a '1 to 1' digital representation of every aspect of the production cycle where every part involved would have a digital representation that not only carry the physical characteristics of the item but also all its information produced over time. To this end, as explained in CHAPTER 2, MES takes a fundamental role in obtaining the real time information required for the DT even be possible.

For instance, a CNC machine would have a digital 3D model for simulation as well as a fully integrated list of all the pieces it produces, data regarding its current level of production, the current wear of its mechanical pieces, all other machines it relates to, history of all the alterations and improvements by which it was affected and many other aspects, all well packaged in an intuitive graphical user interface (GUI) that allows for maximum interaction.

Outside of fiction, we are yet to achieve such level of virtualization. It takes too much time and money to obtain and organize information to such a level of minutia, specially, the aspects that need to be inserted by hand, not to mention the subjectiveness of how this information can be integrated and interacted with. Regardless of that it is useful to identify, within the ideal, the aspects of most importance for this implementation.

Those are:

- The means of virtualization – What sort of information is used to build the virtual items. This includes the metadata and files that are directly attached to the item. In an ideal fashion this would contain all possible information available about the item.

- The means of data input - How this information is being loaded and organized. Ideally this information would be loaded into the system as automatically as possible, be it by means of MES during quality control or through the use of automated input tools like bar code scanners.

## 3.1.這種整合在實際中會是什麼樣

如第 **2** 章中所提到的，**PLM** 的主要想法是管理與產品相關的所有流程中的變更，它主要透過使 虛擬化來實現。這裡的「虛擬化」 詞表 現實世界的項 在數位空間中的表 ，並且正如 們可以想像的那樣，可以透過多個抽象層級來表 真實的物件或過程。因此，關於 **PLM** 虛擬表 必須有多深和**/**或多詳細才能達到其 的，還沒有達成確切的共識。

在理想的世界中，這將是最低的抽象形式，本質上將歸結為數位孿 ，如第 **2** 章所述。這是 產週期各個 的「**1** 對 **1**」數字表 ，其中每個部分所涉及的內容將具有數位表 ，不僅包含該項 的物理特徵，還包含隨著時間的推移產 的所有資訊。為此，如第 **2** 章所述，**MES** 在獲取 **DT** 所需的即時資訊 發揮基礎作 ，甚 是可能的。

例如，數控機床將具有 於模擬的數位 **3D** 模型以及其 產的所有零件的完全整合清單、有關其當前 產 平的數據、其機械零件的當前磨損情況以及與其相關的所有其他機器、受其影響的所有變更和改進的歷史記錄以及許多其他 ，全部都很好地封裝在直覺的圖形使 者介 **(GUI)** 中，可實現最 程度的互動。

除了 說之外，我們還沒有達到這樣的虛擬化程度。獲取和組織如此詳細的資訊需要花費太多的時間和 錢，特別是需要 動插 的 ，更不 說如何整合和互動這些資訊的主觀性了。不管怎樣，在理想情況下確定對於此實施最重要的 是有 的。

那些是：

- 虛擬化的 段－使 什麼樣的資訊來建構虛擬物品。這包括直接附加到項 的元資料和 件。在理想的情況下，這將包含有關該項 的所有可能的可 資訊。

- 資料輸 的 式 **-** 如何載 和組織這些資訊。理想情況下，這些資訊將盡可能 動載 到系統中，無論是在品質控制期間透過 **MES** 還是透過使 條碼掃描器等 動輸 具。

- The means of access – How this information is presented to the users. Although more subjective than the previous aspects this is incredibly important to the way the system is interacted with. How intuitive it is the information availability plays right into the core strengths of PLM. Afterall, everything would be for nothing (even if all else would be perfect) if the only way to interact with the system were a command line interface that would make difficult for the end users to access the information.

- The means of integration - How items and their contained information can interact and benefit from one another, i.e., the integration with other systems and key softwares. E.g., if an item has access to a cad file, there should be no need to fill in the metadata fields by hand. Hoe items can automatically affect other items also plays into this aspect.

- 存取 式－如何將資訊呈現給使 者。儘管 前 的 更主觀，但這對於系統互動的 式常重要。資訊可 性的直覺程度正是 **PLM** 的核 優勢。畢竟，如果與系統互動的唯式是命令列介 ， 命令列介 將使最終 存取資訊變得困難，那麼 切都將毫無意義（即使其他 切都很完美）。

- 整合 式－專案及其所包含的資訊如何互動並相互受益，即與其他系統和關鍵軟體的整合。例如，如果某個項 可以存取 **cad** 件，則無需 動填寫元資料欄位。鋤頭項 可以 動影響其他項 也能發揮作 。

# 4. CHAPTER

# INTRODUCTION TO THE COMPANY AND PRODUCT

As one can imagine, one of the unique aspects of this work is its focus in one specific software solution that tend to be quite flexible in terms of ease of implementation to different sorts of business. This is contrary to most use cases regarding PLM implementation where the business case is the constant and the system is built around it. Nonetheless, in order to evaluate Odoo as a PLM+MES tool, it is important to consider an example. The advantage here is that a fictional company can be picked for this end maximizing the perceived effect of the software during a simulation.

It is considering all those previously mentioned systems that, for the sake of exemplification, the theoretical company was organized in the molds of Industry 4.0. This company is a recently founded small case manufacturing company that uses plastic injection molding as their primary mean of production and uses additive manufacturing and fast prototyping as part of their business strategy. As explained in chapter 2 those are great examples of the path that industry is taking regarding innovation where mass production is becoming slowly less important than product variety and time to market.

In order to maximize the tracking of change, most of its business are based on lower production batches on mainly automated machinery. This company focus in the production of injected plastic products and rely heavily in flexible machinery for setting production and prototyping. Having that in mind, it should be simple enough to simulate continuous improvement of both product and process to the extent of the evaluated software. Since this sort of everchanging production is extremely dependent on information management of all kinds, it must prove to be a perfect base for applied PLM+MES.

In this example the company has already implemented, since its recent foundation, the Odoo software and has taken all the necessary training and steps to its proper use. This allow the removal of the boundaries and limitations that are so common regarding implementation of the PLM+MES system to an already existing business, i.e., dependences on legacy systems administrative resistance to change or integration to old procedures. These are obviously important, but it is not within the scope of this work.

The company aims to produce a completely new product by the end of the year. After doing so, the company improved the process of production for said product. Once there is the need for product improvement, said improvement was performed as well.

# 4. 章節

## 公司及產品介紹

可以想像，這項 作的獨特之處之 是它專注於 個特定的軟體解決 案，該解決 案在易於實施不同類型的業務 往往 常靈活。這與 多數有關 **PLM** 實施的 例相反，在這些 例中，業務案例是不變的，系統是圍繞它構建的。儘管如此，為了評估 **Odoo** 作為 **PLM+MES** 具的能 ，考慮 個範例很重要。這裡的優點是可以選擇 家虛構的公司來實現這 的，從 在模擬過程中最 化軟體的感知效果。

考慮到前 提到的所有系統，為了舉例說明，理論公司是按照 業 **4.0** 的模式組織的。該公司是 家最近成 的 型箱體製造公司，使 塑膠射出成型作為主要 產 式，並使 積層製造和快速原型製作作為其業務策略的 部分。如第 **2** 章所解釋的，這些都是產業在創新所採取的路徑的很好的例 ，在這種創新中， 規模 產逐漸變得不如產品品種和上市時間重要。

為了最 限度地追蹤變化，其 部分業務都基於主要 動化機械上的較低 產批次。該公司專注於注射塑膠產品的 產，並嚴重依賴靈活的機械來進 產和原型製作。考慮到這 點，它應該 夠簡單，可以在評估的軟體範圍內模擬產品和流程的持續改進。由於這種不斷變化的 產極其依賴各種資訊管理，因此它必須成為應 **PLM+MES** 的完美基礎。

在這個例 中，該公司 最近成 以來已經實施了 **Odoo** 軟體，並採取了所有必要的培訓和步驟來正確使 該軟體。這允許消除在現有業務中實施 **PLM+MES** 系統時常 的邊界和限制，即依賴遺留系統管理對更改或整合舊程式的阻 。這些顯然很重要，但不屬於本作的範圍。

該公司的 標是在今年底前 產出全新的產品。在此之後，該公司改進了該產品的 產流程。 旦產品需要改進，也進 改進。

The following diagram (Figure 9) will be taken into consideration as the path of product development and improvement:

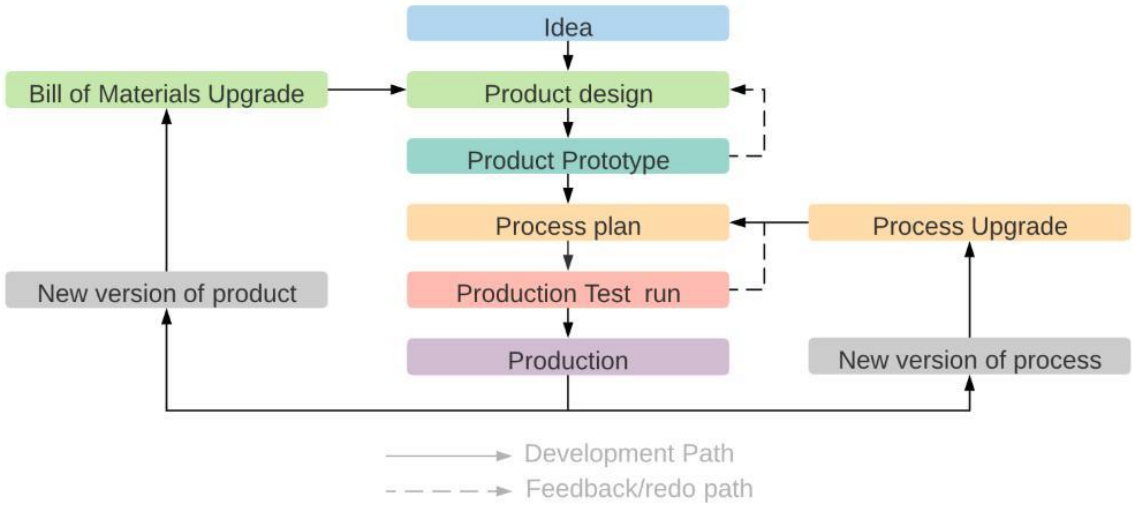


**Figure 9 Development diagram**

This path aims to transmit to the reader an iterative approach towards development and improvement. The idea is followed by a product design for which a cycle of prototyping and redesign takes effect until satisfactory result is achieved. Then a similar cycle takes place regarding the production process. At the end of this stage initial development is done and the actual production can begin.

It is at this point that ways of stablishing the continuous improvement is important. In the case of this company, we are only considering two main types of upgrade paths, those being, product upgrade and process upgrade respectively.

## 4.1. The products and processes

Change and effect are the focus of the PLM+MES implementation as such the subject of said change would ideally be something that could afford a reasonable amount of freedom of design. Although the effects of a well implemented PLM+MES should be substantial even in rigid manufacturing environments, where the change is extremely limited, the system will produce much more perceivable change in an enterprise that thrives in innovation because there will be more opportunities to improve the system and gain feedback.

From the perspective of improvement, if you compare a product that is a result from sheet metal stamping (Figure 10) to an equivalent product that is the result of a CNC milling procedure (Figure 11) it is easy to perceive that the CNC milled product is more welcoming

將考慮下圖（圖 **9**）作為產品開發和改進的路徑：

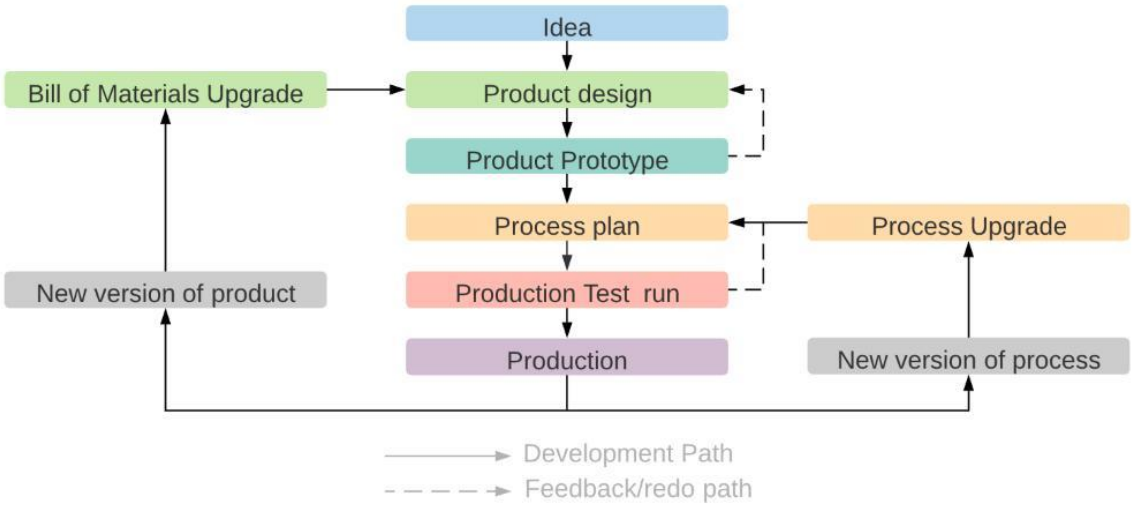


**圖 9** 開發圖

這條路徑旨在向讀者傳達 種開發和改進的迭代 法。這個想法之後是產品設計，原型製作和重新設計的循環 效，直到獲得滿意的結果。然後在 產過程中會發 類似的循環。在此階段結束時，初步開發完成，實際 產即可開始。

正是在這 點上，建 持續改進的 法很重要。就這家公司 ，我們只考慮兩個主要的升級路徑，分別是產品升級和流程升級。

## **4.1.**產品及 藝

變更和效果是 **PLM+MES** 實施的重點，因此理想情況下，上述變更的主題應該能夠提供合理的設計 由。雖然實施良好的 **PLM+MES** 的效果即使在變化極其有限的嚴格製造環境中也應該是顯著的，但該系統將在創新蓬勃發展的企業中產 更明顯的變化，因為將有更多的機會來改進系統並獲得回饋。

從改進的 度來看，如果將鈑 沖壓的產品（圖 **10**）與 **CNC** 銑削加 的同等產品（圖 **11**）進 較，很容易看出 **CNC** 銑削的產品更受歡迎

to upgrades. While the stamping is low cost (by comparison) it depends on heavy high precision metal dyes that are extremely expensive to produce. This means that the cost of enacting change to it is much higher and thus the effect of a system that thrives on tracking change becomes limited.

**Figure 10 Example of stamped AK74 pattern rifle receiver (Brownnells.com)**

**Figure 11 Example of milled AK74 pattern rifle receiver (sharpsbros.com)**

In the case of this fictional company, it has been determined that the best way to exemplify the PLM+MES effects would be to have products designed around plastic injection molding. It might seem unintuitive at first to consider this manufacturing procedure, like the stamping procedure previously described, since it too depends on high precision molds during production. However, the main differences between the two is regarding ease of prototyping and the cost of upgrading.

Injection molding is a broad and complex field of engineering that involves a huge variety of materials and methods, little of which is of the concern of this work. It is however relevant to point out that for the most part, the pressures involved in the injection molding are one order of magnitude lower than the when we are dealing with steel; softer materials can be

來升級。雖然沖壓成本較低（相 之下），但它依賴於 產成本極其昂貴的重型 精度 屬染料。這意味著對其進 更改的成本要 得多，因此依靠追蹤更改 蓬勃發展的系統的效果變得有限。

**圖 10** 沖壓 **AK74** 型步槍機匣範例

（**Brownnells.com**）

**圖 11** 銑削 **AK74** 型步槍機匣範例

（**sharpsbros.com**）

就這家虛構的公司 ，我們確定體現 **PLM+MES** 效果的最佳 式是圍繞塑膠射出成型設計產品。乍 看，考慮這種製造過程（如前 描述的沖壓過程）似乎不直觀，因為它在產過程中也依賴 精度模具。然 ，兩者之間的主要區別在於原型設計的難易度和升級成本。

射出成型是 個廣泛 複雜的 程領域，涉及各種各樣的材料和 法，但本 作只涉及其中很少 部分。然 ，需要指出的是，在 多數情況下，射出成型中涉及的壓 我們處理鋼時的壓 低 個數量級；較軟的材料可以

used on their molds like CNC milled aluminum. At the same time, new advancements in the field of additive manufacturing have made possible to prototype plastic parts with much closer physical characteristics to the end result of a injected piece. Sometimes even prototype molds (Figure 12) can be used for a lower volume test runs during process upgrades.

**Figure 12 Example of injection mold made using a 3D printer (thefabricator.com)**

Additive manufacturing has become an incredible tool for ultra-flexible production. This mindset of continuous improvement, especially when regarding prototyping and iterative design, is a hallmark of the lean mentality that is so relevant in the modern industry.

As mentioned in the previous section, in this case study it is considered the creation of a new product and its production process by the fictional company. This product consists in a plastic small form factor computer case, composed of 3 different parts (Figure 13) that are expected to be designed and prototyped considering combination of additive manufacturing and CNC milling towards a plastic injection molding production.

於他們的模具，如 **CNC** 銑削鋁。同時，積層製造領域的新進步使得塑膠零件原型成為可能，其物理特性與注射件的最終結果更加接近。有時，甚 原型模具（圖 **12**）也可 於製程升級期間的 批量試運 。

圖 **12** 使 **3D** 列印機製作的射出成型模具範例

（**thefabricator.com**）

增材製造已成為超靈活 產的不可思議的 具。這種持續改進的 態，特別是在原型設計和迭代設計 ，是與現代 業密切相關的精益 態的標誌。

如上 節所提到的，在本案例研究中，它被視為虛構公司的新產品的創造及其 產過程。該產品由 個塑膠 型電腦機殼組成，由 **3** 個不同的部件組成（圖 **13**），預計在設計和原型製作時考慮將增材製造和 **CNC** 銑削相結合進 塑膠射出成型 產。

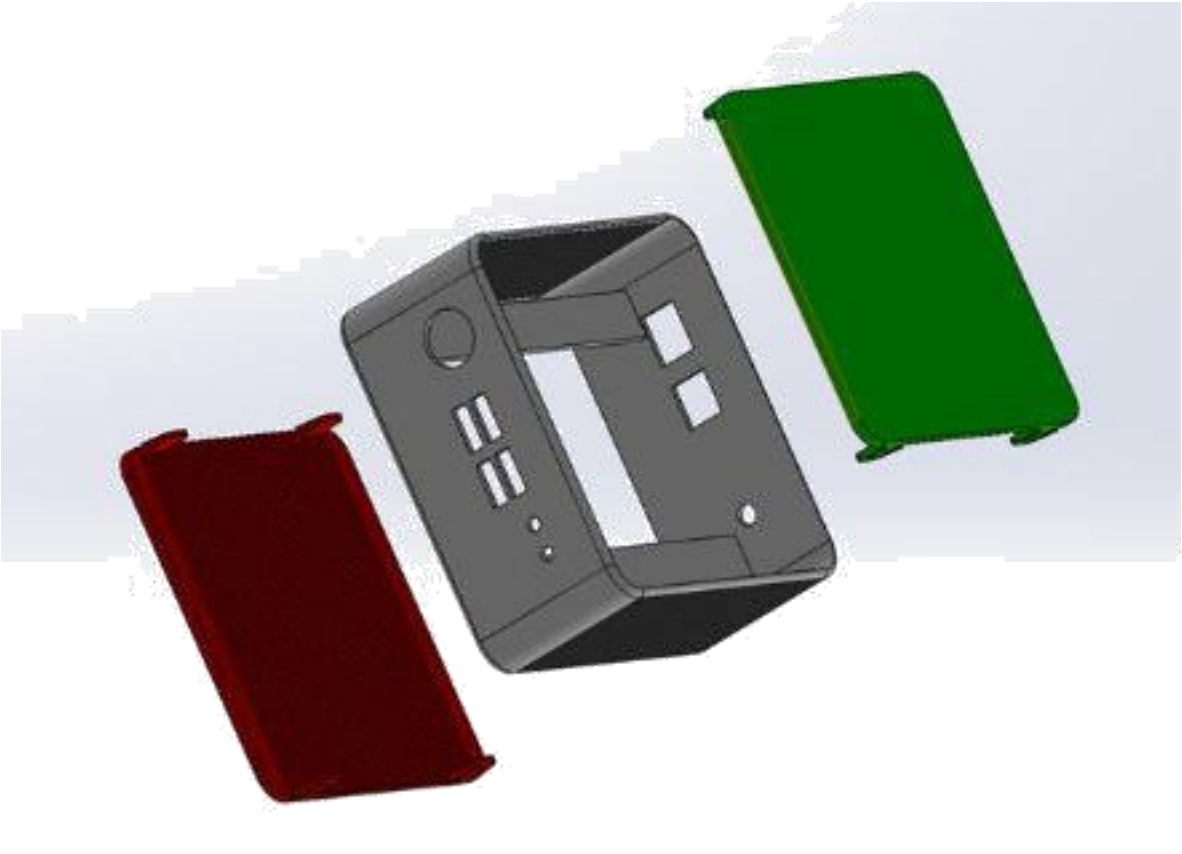

**Figure 13 3D exploded view of the theoretical product**

### 4.1.1. Part A

PART-A (Figure 14) is the core structure of the computer case. It is expected to comport all the pieces necessary for the proper function of the small form factor computer in question. To this end a raw material A was selected to be Acrylonitrile Butadiene Styrene (ABS) this is an opaque thermoplastic polymer and an engineering grade plastic. It is commonly used to produce electronic parts such as phone adaptors, keyboard keys and wall socket plastic guards.

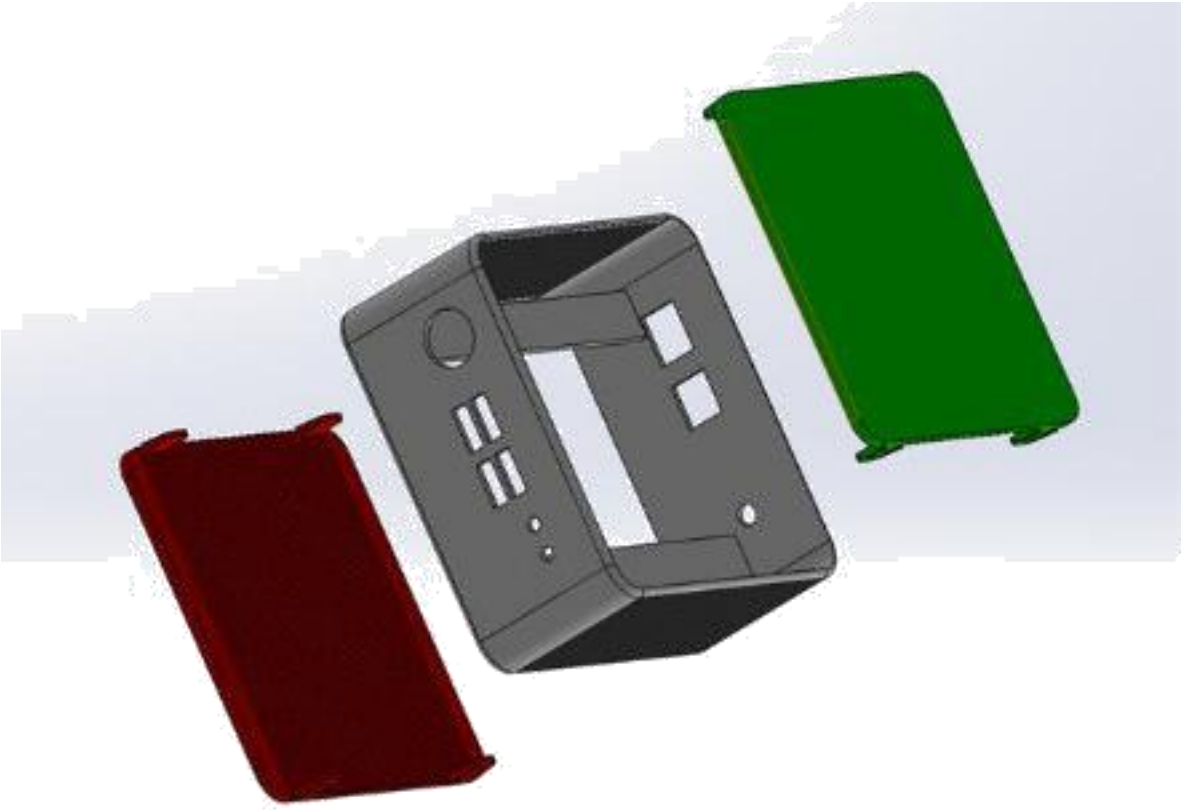

圖 **13** 理論產品 **3D** 分解圖

### **4.1.1. A** 部分

**PART-A**（圖 **14**）是電腦機殼的核 結構。它預計將配備所討論的 型計算機正常運所需的所有部件。為此，原料 **A** 被選為丙烯腈丁 烯苯 烯 **(ABS)**，這是 種不透明的熱塑性聚合物和 程級塑膠。它通常 於 產電 零件，例如電話適配器、鍵盤按鍵和牆壁插座塑膠防護罩。

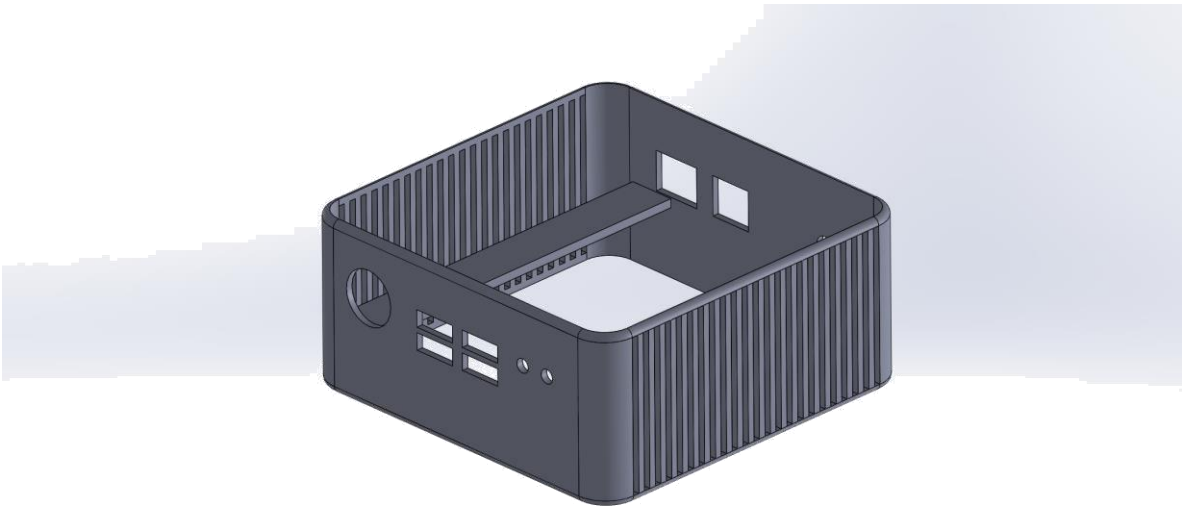

**Figure 14 Isometric view of Part A**

The main reasons for choosing this material specifically are its toughness, its good dimensional stability (resistance to change dimensions after cooling), its high impact resistance and surface hardness. Finally, it is also commonly available in the form of 3D printing filament for extrusion 3D printers which should prove to be quite useful during prototyping.

### 4.1.2. Parts B and C

Parts B and C are lids that should snap into place, closing the system. These are very simple pieces and require a certain level of elasticity so it can deform to assure a screwless assembly. These two identical parts are going to be made with Thermoplastic Polyurethane (TPU), because of its elastic nature and great tensile and tear strength. This sort of polymer is often used to produce parts that demand a rubber-like elasticity. TPU performs well at high temperatures and is commonly used in power tools, cable insulations and sporting goods. Finally, TPU is also available in the form of filament for 3D printers which, for the simulation, will be used for prototyping.

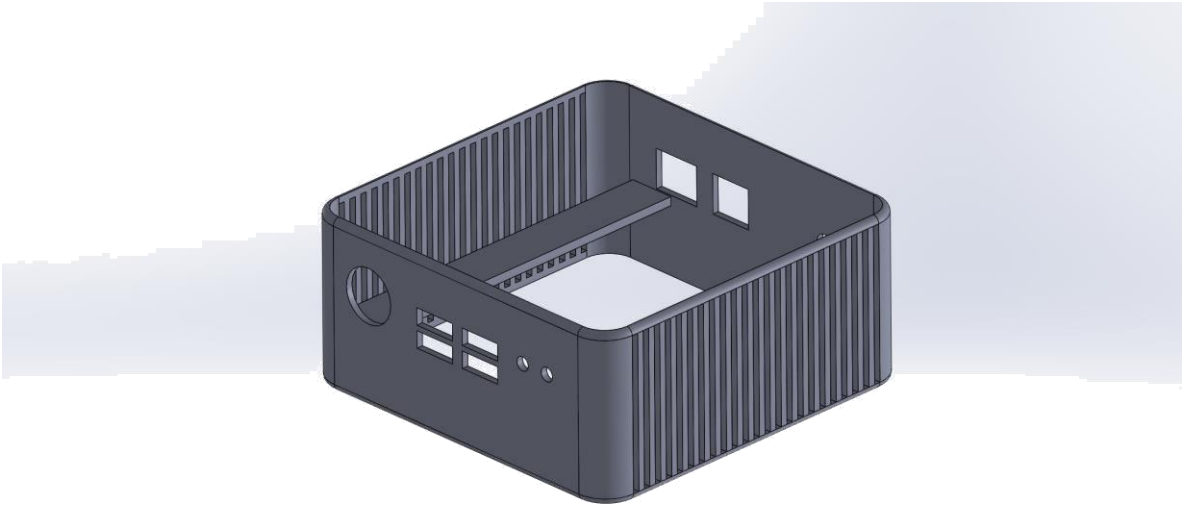

圖 **14 A** 部分等距視圖

特別選擇這種材料的主要原因是其韌性、良好的尺 穩定性（冷卻後不易改變尺）、 抗衝擊性和表 硬度。最後，它通常也以 於擠出 **3D** 列印機的 **3D** 列印絲的形式提供，這在原型製作過程中應該 常有 。

**4.1.2. B** 部分和 **C** 部分

**B** 和 **C** 部分是蓋 ，應卡 到位，關閉系統。這些都是 常簡單的零件，需要 定程度的彈性，以便可以變形以確保無螺絲組裝。這兩個相同的部件將由熱塑性聚氨酯 **(TPU)** 製成，因為它具有彈性、拉伸強度和撕裂強度。這種聚合物通常 於 產需要類似橡膠彈性的零件。 **TPU** 在 溫下表現良好，常 於電動 具、電纜絕緣材料和體育 品。最後，**TPU** 還可以以細絲的形式 於 **3D** 列印機， 於模擬， 於原型製作。

**Figure 15 Parts B and C**

### 4.1.3. Molds

Ideally all molds should be made of steel, for longevity of the mold and product quality. That being said, the injected plastics that are being selected for all parts are not so pressure dependent and their forms are not so complex, so it is assumed that aluminum molds made with a precision CNC machining should suffice to produce said parts.

It is also assumed that all molds are simple enough to be prototyped using 3D printing. Although this is not always true, it was determined representative enough for this simulation. The type of material used in those prototypes is high temperature resign cured using an SLA 3DPrinter. Additionally, the mold will be considered the main physical aspect to be developed when regarding the production process because it something that directly affects the production as well as something that can be produced in house and tracked as a product would.

## 4.2. What is analized during the simulation

Taking into consideration the diagram, shown in Figure 9, as well as the main aspects of a successful integration of PLM and MES as described in the section 3.1, this experiment aims to produce commentary regarding the following relevant questions in Table 1.

圖 **15 B** 和 **C** 部分

### **4.1.3.**模具

理想情況下，所有模具應由鋼製成，以確保模具的使 壽命和產品品質。話雖如此，為所有零件選擇的注射塑膠並不那麼依賴壓 ， 且它們的形狀也不是那麼複雜，因此假設 精密 **CNC** 加 製造的鋁模具應該 以 產所述零件。

也假設所有模具都 夠簡單，可以使 **3D** 列印來製作原型。儘管這並不總是正確的，但它對於本次模擬來說具有 夠的代表性。這些原型中使 的材料類型是使 **SLA 3D** 列印機進 溫重新固化的。此外，在 產過程中，模具將被視為要開發的主要物理 ，因為它直接影響 產，並且可以像產品 樣在內部 產和追蹤。

## **4.2.**模擬過程中分析的內容

考慮到圖 **9** 所 的圖表，以及第 **3.1** 節中描述的 **PLM** 和 **MES** 成功整合的主要 ，本實驗旨在對錶 **1** 中的以下相關問題進 評論。

**Table 1 Summary of questions to be answered**

| Category | Questions |
|---|---|
| How does the software deals with items? | Are all aspects of the product lifecycle represented? |
| | How well are each of those items represented? |
| How easy it is to create a brand-new product? | How the product is depicted |
| | How does the product integrate and reference relevant files? |
| | Does changing one affects the other? |
| How easy it is to create a brand-new production process? | How the process is depicted? |
| | How does the process integrate and reference the product it produces? |
| | Does changing one affects the other? |
| How easy is to improve an existing product | How easy it is to update its metadata |
| | How easy it is to determine the effects of the change |
| | How does the software deals with different product revisions? |
| How easy it is to improve an existing production process | How easy it is to update its metadata |
| | How easy it is to determine the effects of the change |
| | How does the software deals with different production process revisions? |
| How easy is to find data related to product or process? | How easy is find production numbers? |
| | How does Odoo generate performance data? |
| | How does the software present performance change as a result of a upgrade? |

表 **1** 需回答的問題總表

| 類別 | 問題 |
| --- | --- |
| 軟體如何處理專案？ | 是否代表了產品 命週期的所有 向？<br>這些項 的表現如何？ |
| 創造 個全新的產品有多容易？ | 產品的描述 式<br>產品如何整合和引 相關 件？<br>改變 個會影響另 個嗎？ |
| 創造 個全新的產品是多麼容易 產過程？ | 其過程是如何描述的？<br>該流程如何整合和參考其 產的產品？<br>改變 個會影響另 個嗎？ |
| 改進現有產品有多容易 | 更新元數據有多容易<br>確定變更的影響有多容易<br>軟體如何處理不同的產品版本？ |
| 改進現有 產流程有多容易 | 更新元數據有多容易<br>確定變更的影響有多容易<br>軟體如何因應不同的 產流程修訂？ |
| 尋找與產品或流程相關的數據有多容易？ | 要找出 產編號有多容易？<br>**Odoo** 如何產 效能數據？<br>升級後軟體的效能有何變化？ |

# 5. CHAPTER

# THE ODOO SOFTWARE

## 5.1. Introduction to the Odoo software

Odoo is a commercial business management software with strong ties to the open source community. Initially started as open source ERP software becoming well received as an affordable and intuitive package that thrived on integration and expandability. Since then, as the company experienced accelerated growth, it shifted their business model to include an enterprise paid version as well as an online service.

As mentioned in the section 2.2, modern ERP systems are usually modular and, in the case of Odoo, this modularity is particularly evident due to the incredible amount of expansion provided by community developed modules as well as company developed modules that are highly integrated. This extendibility is what makes this software so relevant to the topic of PLM+MES integration since there are present modules for PLM as well as noticeable MES functionalities within their manufacturing modules.

Within the scope of this thesis, the objective is to utilize this software on the management of the previously mentioned fictional company and draw conclusions regarding how effective the integration of PLM and MES is already present within this system.

### 5.1.1. How it works

The software can be installed in most x86 computers and it supports several operating systems including windows and all the main Linux distributions.

Ideally, the Odoo software is installed in a computer connected to a local area network and starts a SQL database that holds all the necessary information and files produced by the business (Figure 16). Said computer works essentially as a server and accessed via a browser by the other machines present in the network. This computer can be a dedicated server or a working desktop in use, but it is important to remember that it must remain ON and connected throughout the entire time the software is required to function.

# 5. 章節

# ODOO 軟體

## 5.1. Odoo 軟體簡介

**Odoo** 是 款與開源社群有著密切聯繫的商業企業管理軟體。最初是作為開源 **ERP** 軟體開始的，後來作為 種經濟實惠且直觀的軟體包 廣受好評，並在整合和可擴展性 蓬勃發展。從那時起，隨著公司經歷加速成 ，它改變了他們的業務模式，包括企業付費版本和線上服務。

如同 **2.2** 節中所提到的，現代 **ERP** 系統通常是模組化的，就 **Odoo** ，這種模組化尤其明顯，因為社群開發的模組以及公司開發的 度整合的模組提供了令 難以置信的擴展量。這種可擴展性使得該軟體與 **PLM+MES** 整合主題如此相關，因為其製造模組中存在 **PLM** 模組以及顯著的 **MES** 功能。

在本論 的範圍內， 標是利 該軟體來管理前 提到的虛構公司，並就該系統中 **PLM** 和 **MES** 整合的有效性得出結論。

### 5.1.1.怎麼運作的

該軟體可以安裝在 多數 **x86** 電腦上，並且 援多種作業系統，包括 **Windows** 和所有主要的 **Linux** 發 版。

理想情況下，**Odoo** 軟體安裝在連接到區域網路的電腦中，並啟動 **SQL** 資料庫，該資料庫保存企業產 的所有必要資訊和檔案（圖 **16**）。所述電腦本質上是作為伺服器 作，並由網路中存在的其他機器透過瀏覽器存取。這台電腦可以是專 伺服器或正在使 的 作桌 ，但請務必記住，它必須在軟體需要運 的整個過程中保持開啟和連接狀態。